Topic 7: Solved Problems

Step-by-step solutions of all 2 problems of the magnetic part of the official sheet (Problems T6andT7_Electric and Magnetic Properties, problems 11 and 12: the hysteresis loops of soft and hard magnetic materials and the effect of the microstructure on a ferromagnetic material, from Session 15). Problems 1 to 10 of the sheet are electrical and belong to Topic 6. The sheet prints no solution for these two, so both are derived here.

`Problem 11` `Problems T6andT7_Electric and Magnetic Properties.pdf #11` `Soft and hard magnets · B-H loop`

Hysteresis Loops of Soft and Hard Magnetic Materials

`Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 3`

> *11. Draw the hysteresis curve of a soft magnetic material and a hard magnetic material and indicated in both cases the most significant features. What are the main differences between these two materials?*
>
> *The sheet prints no solution for this problem.*

`HYPOTHESES & PARAMETERS`

Hypotheses

1. **Saturated loop, quasi-static.** The loop is the one obtained by cycling the field slowly between $+H_{max}$ and $-H_{max}$, with H_{max} large enough to reach saturation. The drawing is a schematic, as the statement asks for a sketch and gives no numbers.
2. **Features to indicate.** The features of the slides are the saturation induction B_s (saturation magnetization M_s), the remanence B_r, the coercivity H_c and the permeabilities μ_i and μ_{max} [Session 15 Slides 16-17].
3. **Soft and hard.** The two classes are separated by the coercivity, $H_c = 10\,000$ A/m [Session 15 Slide 18].
4. **Illustration with data.** The numbers come from the tables of Slide 24 of Session 15: Table 20-4 (soft materials, with H_c in A/m) and Table 20-6 (hard materials, with $\mu_0 H_c$ in tesla). $\mu_0 = 1.257 \times 10^{-6}$ H/m [Session 15 Slide 4].

Data

Soft material	$\mu_{r,max}$	H_c in A/m	B_r in T
Ingot iron (99.8 % Fe)	5000	80	0.77
Fe-3 % Si, grain-oriented	50 000	7	1.20
Superalloy (Fe-5 % Mo-80 % Ni)	450 000	0.4	not printed

Hard material	$\mu_0 M_r$ in T	$\mu_0 H_c$ in T
Co-steel (Fe-Co)	1.07	0.02
AlNico-5 (Fe-Co-Al-Ni)	1.05	0.06
Ferrite ($BaFe_{12}O_{19}$)	0.42	0.31
Sm-Co ($SmCo_5$)	0.87	0.80
Nd-Fe-B ($Nd_2Fe_{14}B$)	1.23	1.21

Unknown: the two loops with their features, and the differences between the materials.

MODEL & STRATEGY

The loop comes from the motion of the Bloch walls. In a soft material the walls move easily, so a small field is enough to magnetize the material and to demagnetize it again. In a hard material the walls are difficult to move, so a large field is needed in both directions [Session 15 Slide 18].

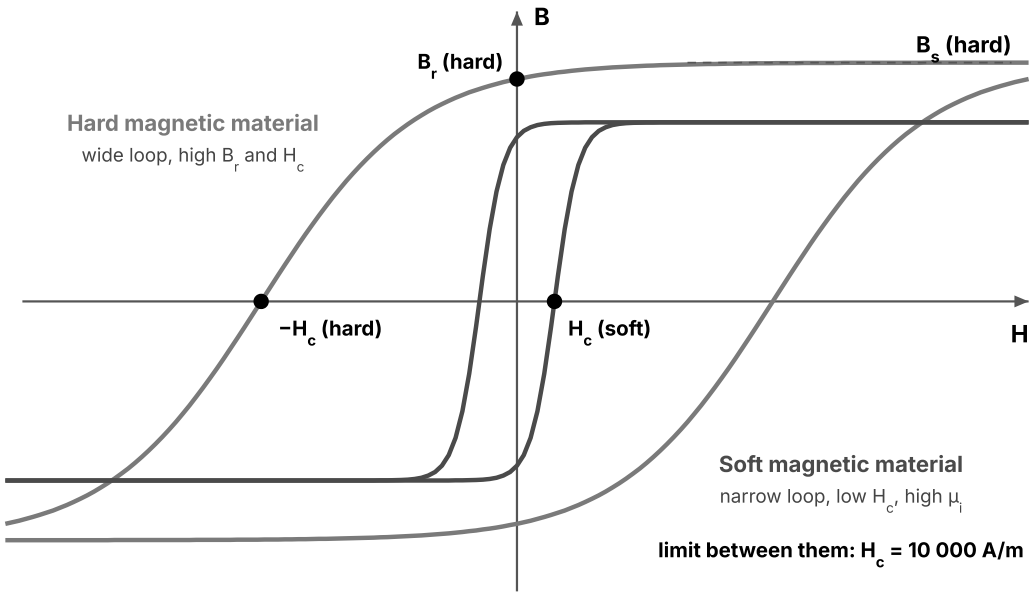


Figure 1. B-H hysteresis loops: a soft material has a narrow loop (low coercivity Hc, high initial permeability), a hard material a wide one (high remanence Br, saturation Bs and Hc); the slide fixes the limit at Hc = 10 000 A/m (schematic).

Strategy.

1. Follow the loop step by step and mark where each feature appears.
2. Put numbers on the features with the tables of Slide 24, converting the hard-material coercivity to A/m so that both classes are compared with the same unit and with the limit of 10 000 A/m.
3. Compare the two classes feature by feature.

STEP-BY-STEP CALCULATION

1. Path of the loop [Session 15 Slides 16-17]

- Starting from the demagnetized state ($H = 0$, domains at random) the initial curve rises to saturation: $B = B_s$.
- Reducing H to zero does not bring B back to zero: the induction left is the **remanence** B_r.
- A field in the opposite direction, of magnitude H_c, brings B to zero: the **coercivity**.
- Going on to $-H_{max}$ gives $-B_s$, and the return branch is the mirror image through the origin.

The slope of the initial curve is the initial permeability μ_i, and the line from the origin tangent to the curve gives the maximum permeability μ_{max}.

2. Features on each loop

Feature	Where	Soft material	Hard material
μ_i, μ_{max}	slope of the initial curve	high	low μ_i
H_c	crossing of the H axis	low, below 10 000 A/m	high, above 10 000 A/m
B_r	crossing of the B axis	printed as a feature, not as a rule	high, printed on Slide 18
B_s	plateau	reached with a small field	reached only with a large field
Loop area	enclosed by the loop	small	large

3. Hard-material coercivity in A/m

From $\mu_0 H_c$ in tesla, with $\mu_0 = 1.257 \times 10^{-6}$ H/m:

$$H_c = \frac{\mu_0 H_c}{\mu_0}$$

For Nd-Fe-B:

$$H_c = \frac{1.21\ \mathrm{T}}{1.257 \times 10^{-6}\ \mathrm{H/m}} = 9.626 \times 10^{5}\ \mathrm{A/m}$$

The same division for the others:

Hard material	H_c in A/m	Above 10 000 A/m?
Co-steel	1.59×10^{4}	yes
Alnico-5	4.77×10^{4}	yes
Ferrite	2.47×10^{5}	yes

Hard material	H_c in A/m	Above 10 000 A/m?
Sm-Co	6.36×10^5	yes
Nd-Fe-B	9.63×10^5	yes

Ratio of the strongest hard material to the soft ones of the table:

$$\frac{9.626 \times 10^5 \text{ A/m}}{80 \text{ A/m}} = 1.2 \times 10^4 \qquad \frac{9.626 \times 10^5 \text{ A/m}}{7 \text{ A/m}} = 1.4 \times 10^5$$

4. Energy lost per cycle (extension, order of magnitude)

The energy dissipated per unit volume in one cycle is the area enclosed by the loop, $W = \oint H\, dB$ (this relation is not printed on the slides). For an idealized rectangular loop of width $2H_c$ and height $2B_r$ the area is

$$W \approx 2H_c \times 2B_r = 4\, H_c\, B_r$$

For the grain-oriented Fe-3 % Si and for Nd-Fe-B:

$$W_{soft} \approx 4 \times 7 \text{ A/m} \times 1.20 \text{ T} = 33.6 \text{ J/m}^3 \qquad W_{hard} \approx 4 \times 9.626 \times 10^5 \text{ A/m} \times 1.23 \text{ T} = 4.74 \times 10^6 \text{ J/m}^3$$

The ratio is about 1.4×10^5, the same as the ratio of coercivities, because the two remanences are similar. This is why the soft materials are the ones used under alternating fields, where the cycle repeats many times per second, and the hard ones are used as permanent magnets, where it never repeats [Session 15 Slides 18-19].

5. Main differences

	Soft magnetic material	Hard magnetic material
Bloch walls	easy to move	difficult to move
Loop	narrow	wide
Coercivity	$H_c < 10\,000$ A/m	$H_c > 10\,000$ A/m
Permeability	high initial permeability	low initial permeability
Losses per cycle	low	high
Demagnetization	easy	resistant

	Soft magnetic material	Hard magnetic material
Use	cores of transformers, generators and motors; materials under alternating fields	permanent magnets

RESULT & CHECKS

Checks

- **Units of H_c.** $[\mu_0 H_c]/[\mu_0] = \mathrm{T}/(\mathrm{H/m}) = (\mathrm{Wb/m^2})/(\mathrm{Wb/(A \cdot m)}) = \mathrm{A/m}$.
- **Units of the energy.** $[H\,B] = (\mathrm{A/m})(\mathrm{T}) = \mathrm{A \cdot T/m}$, and with $\mathrm{T} = \mathrm{N/(A \cdot m)}$ this is $\mathrm{N/m^2} = \mathrm{J/m^3}$, an energy per unit volume.
- **The limit of the slide.** All the soft materials of the table are below 10 000 A/m (the highest is 100 A/m for the low-carbon steel of Table 20-4) and all the hard ones above it. Co-steel, at 1.59×10^4 A/m, is the closest to the limit.
- **Limit $H_c \to 0$.** The loop collapses into a single line with no area and no loss, the ideal soft material. For a very large H_c the loop is so wide that the material keeps its magnetization against the fields it meets in use, the ideal permanent magnet.

Interpretation

The decisive difference is the coercivity, and everything else follows from the mobility of the Bloch walls: the width of the loop, the losses and the use. The remanence alone does not separate the classes. The grain-oriented Fe-3 % Si has $B_r = 1.20$ T, more than the ferrite magnet (0.42 T) and about the same as Nd-Fe-B (1.23 T). Slide 18 lists a high remanence among the properties of the hard materials; in the data of Slide 24 it is the coercivity that is orders of magnitude apart (7 to 100 A/m against 10^4 to 10^6 A/m), not the remanence.

Check against the official solution

Item	Official	Computed	Comment
Loops and features	not printed on the sheet	B_s, B_r, H_c, μ_i, μ_{max} on both loops	consistent with Session 15 Slides 16-18
Differences	not printed on the sheet	coercivity, loop width, permeability, losses and use	consistent with Session 15 Slides 18-19; the energy per cycle and the numbers of Slide 24 are an extension

Problem 12 | Problems T6andT7_Electric and Magnetic Properties.pdf #12 | Microstructure · ferromagnetic properties

Effect of Grain Size, Impurities and Porosity on a Ferromagnetic Material

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 3

12. Specify in the following table how the increase in the following microstructural parameters affect the listed properties of a ferromagnetic material

	Grain size	***Impurities***	***Porosity***
Hysteresis losses			
Conductivity			
Permeability			
Coercitivity			
Saturation Magnetization			
Curie Temperature			

The sheet prints no solution for this problem.

HYPOTHESES & PARAMETERS

Hypotheses

1. **One parameter at a time.** Each column is an increase of that parameter with the composition of the ferromagnetic material and the other parameters kept constant.
2. **Domain walls control the loop.** A larger grain gives the Bloch walls more mobility, and inclusions, dislocations and pores reduce it. With a larger mobility the permeability grows and the coercivity and the remanence fall; with a smaller mobility the opposite happens [Session 15 Slide 21].
3. **Impurities and porosity as obstacles.** The slide groups inclusions, dislocations and pores as obstacles to the walls, so impurities (as particles of a second phase) and pores act in the same way on the loop. The effect of dissolved atoms is the same in direction (extension).
4. **Intrinsic properties.** The saturation magnetization is constant for a material of a given composition [Session 15 Slide 20]. The Curie temperature is the temperature at which the saturation magnetization falls to zero, and the table of Slide 23 gives a different value for each element; both are fixed by the composition and the bonding, not by the microstructure (the link between T_c and the composition is an extension).
5. **Conductivity.** The electrical resistivity grows with the structural imperfections (grain boundaries, dislocations) and with the impurities [Session 13 Slides 16-17]; the effect of pores is an extension.
6. **Hysteresis losses.** The energy lost per cycle is the area of the loop (extension, as in Problem - T7-11 Hysteresis Loops of Soft and Hard Magnetic Materials): a wider loop (higher H_c) means higher losses.

Unknown: the effect (increase, decrease or none) of each parameter on each of the six properties.

MODEL & STRATEGY

Two chains separate the properties. The first goes from the microstructure to the mobility of the Bloch walls, and from the mobility to the permeability, the coercivity and the area of the loop. The second contains the properties that the microstructure does not change at all: the saturation magnetization and the Curie temperature, set by the composition. The conductivity follows its own chain, the scattering of the conduction electrons.

Parameter	Mobility of the Bloch walls	Permeability	Coercivity	Loop area (losses)
Larger grain size	increases	increases	decreases	decreases
More impurities, pores	decreases	decreases	increases	increases

Strategy.

1. Fill the first chain from Slide 21.
2. Decide the conductivity from the scattering of the electrons, with Session 13.
3. State why the saturation magnetization and the Curie temperature do not change with the microstructure.
4. Collect the answers in the table of the statement.

STEP-BY-STEP CALCULATION

1. Grain size [Session 15 Slide 21]

A larger grain size increases the mobility of the Bloch walls, because there are fewer grain boundaries to stop them. So:

- permeability μ: **increases**;
- coercivity H_c: **decreases** (and the remanence B_r too);
- hysteresis losses: **decrease**, since a narrower loop has a smaller area;
- conductivity: **increases**, because the grain boundaries are structural imperfections that scatter the electrons and raise the resistivity [Session 13 Slides 16-17].

2. Impurities [Session 15 Slide 21]

Inclusions of another phase are obstacles that pin the walls, and the mobility decreases. So:

- permeability: **decreases**;
- coercivity: **increases** (and the remanence too);
- hysteresis losses: **increase**;
- conductivity: **decreases**, because foreign atoms and particles raise the residual resistivity ρ_i [Session 13 Slide 16].

3. Porosity [Session 15 Slide 21]

Pores are listed on the slide with the inclusions, so the loop changes as for the impurities:

- permeability: **decreases**;
- coercivity: **increases**;
- hysteresis losses: **increase**;
- conductivity: **decreases** (extension): the pores remove conducting section and scatter the electrons, so the same direction as for the impurities.

4. Saturation magnetization [Session 15 Slide 20]

The saturation magnetization is constant for a material of a given composition: it is reached when all the dipoles of the solid are aligned, and the number and the intensity of the dipoles are fixed by the composition. Therefore:

- grain size: **no effect**;
- impurities: **no effect** while they are only inclusions; if an impurity dissolves and changes the composition the value changes (extension);
- porosity: **no effect** on the material itself. Per unit of total volume the pores only dilute the magnetization (extension).

5. Curie temperature [Session 15 Slide 23]

T_c is the temperature at which the saturation magnetization falls abruptly to zero. Each material has its own value (Fe 1043 K, Co 1404 K, Ni 631 K, Gd 289 K), which depends on the elements, not on the size of the grains or on the pores. Grain size and porosity have **no effect**. Impurities have **no effect** as inclusions; dissolved elements that change the composition can shift it (extension).

6. Result

Property	Grain size increases	Impurities increase	Porosity increases
Hysteresis losses	decrease	increase	increase
Conductivity	increases	decreases	decreases
Permeability	increases	decreases	decreases
Coercivity	decreases	increases	increases
Saturation magnetization	no effect	no effect as inclusions	no effect
Curie temperature	no effect	no effect as inclusions	no effect

RESULT & CHECKS

Checks

- **Consistency of the table.** Permeability and coercivity move in opposite directions in every column, as the slide says (μ up with H_c and B_r down for a larger grain, and the reverse for the obstacles). The losses follow the coercivity.
- **Impurities and porosity act alike.** Slide 21 puts inclusions, dislocations and pores in the same line, and the table gives the same entries for both columns, except for the properties that depend on the composition.
- **Limits.** For no obstacles (a pure, dense, coarse-grained material) the walls move freely: the highest permeability and the lowest coercivity and losses. For a very porous material the loop widens and the section that carries the flux and the current shrinks.

Interpretation

One rule explains the table: whatever stops the Bloch walls raises the coercivity and the losses and lowers the permeability, and whatever frees them does the opposite. The saturation magnetization and the Curie temperature are properties of the composition, so no change in the microstructure alone moves them. The same rule is used on purpose in the two classes of Problem T7-11: soft magnetic materials are made pure, dense and with large grains, and hard magnetic materials are made with obstacles (fine grains, precipitates), which is an extension of the slide. A higher conductivity also raises the currents induced in a core under alternating fields, which adds to the losses; this is an extension, and the table asks only for the effects listed.

Check against the official solution

Item	Official	Computed	Comment
All 18 entries of the table	not printed on the sheet	table of Section 3	permeability, coercivity: Session 15 Slide 21; saturation magnetization: Slide 20; conductivity: Session 13 Slides 16-17; the entries marked extension (porosity and conductivity, effect of composition on M_s and T_c, losses as loop area) are not printed on the slides